

ENERGY INPUT

Protected matched 1S LiPo cells in opposite bays
Charge dock pads -> MCP73831, 100-150 mA
VBAT monitor to nRF ADC

POWER TREE

TPS7A2033 -> +3V3 logic
TPS61023 -> switched +5V_LED
TPS22918 load switch and firmware enable

CONTROL

Raytac MDBT50Q-1MV2 / nRF52840
BLE data, frame renderer, eight LED streams
SWD pogo pads; watchdog and brownout

SENSORS

DRV5033 Hall index
BMI270 gyro/accelerometer
Angle estimator corrects hand-speed var

TOP DISPLAY

4 independent chains x 20 SK6805-EC15
5 V data via first 74AHCT125
Radial POV resolution: 20 pixels

CYLINDRICAL TICKER

4 separate chains x 12 SK6805-EC10
Second 74AHCT125; replaceable 0.6 mm boards
Vertical ticker resolution: 12 pixels

1S BATTERY DOMAIN

VBAT range 3.0-4.2 V
 Only protected cells; matched pair, opposite placement
 Fuse/current-limited interconnect required in enclosure
 No charging while spinning

3.3 V LOGIC

TPS7A2033 LDO, 300 mA
 10 uF input/output plus 100 nF at each IC
 nRF52840, BMI270, Hall and level-shifter inputs
 Expected logic load below 50 mA

5 V LED RAIL

TPS61023 synchronous boost, 2.2 uH shielded inductor
 TPS22918 controlled load switch
 4 x 100 uF bulk at arm roots
 1.5 A design target; thermal validation mandatory

FIRMWARE CURRENT CONTRACT

128 x 5 mA pixels can exceed the practical continuous budget at unrestricted white.
 Hard cap global brightness to 35%; default moving graphics target 20-25% average light.
 Reject frames that exceed the measured rail budget; ramp power on and off; log undervoltage.
 Runtime estimates are preliminary and must be replaced with measured current from first article.

nRF52840 MODULE

BLE receives compact market-data frames
Eight hardware-timed 800 kbit/s outputs
Frame phase is reset by Hall index
Gyro interpolation handles nonuniform hand spin

LEVEL SHIFT

2 x 74AHCT125 powered from +5V_LED
3.3 V input-high compatible
Eight buffers: four top plus four tip
Outputs disabled until LED rail is stable

SENSOR INTERFACE

DRV5033 digital Hall output to interrupt GPIO
BMI270 over I2C; optional SPI footprint reassignment
Place IMU near rotation axis
Debounce and reject impossible period changes

TIMING TARGET

Typical hand spin assumption for firmware testing:
300-900 RPM (5-15 revolutions/s).

A 20-pixel top-chain update is about 0.60 ms; four
arms draw four angular lines in parallel.

A 12-pixel tip-chain update is about 0.36 ms and is
scheduled independently from the top image.

Final angular resolution and frame stability must be
measured; marketing renders are not acceptance
evidence.

ARM A / B / C / D (IDENTICAL)

nRF GPIO -> U2 -> 20 x SK6805-EC15 top LEDs

Separate nRF GPIO -> U9 -> 12 x SK6805-EC10 tip LEDs

5 V and GND carried on wide local rails

Return DOUT test pad allows walking-one verification

TIP BOARD INTERFACE

Pad 1: +5V_LED

Pad 2: GND

Pad 3: independent TIP_x_DIN stream

Pad 4: DOUT / test return

Mechanical enclosure guides carry impact load; solder tabs are not structural members

ASSEMBLY CHECKS

Confirm pin-1 orientation against manufacturer datasheet before stencil release.

AOI must verify 80 top and 48 tip LEDs for rotation and solder bridging.

Program walking-one RGB test before enclosing the assembly.

Balance correction comes after all four tip boards, cells and diffuser parts are installed.